

Worksheet 2.2

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Branch: MCA (AI & ML)

Semester: II

Subject Name: Advance Internet Programming

UID: 25IMC13004

Section/Group: MAM-1 (A)

Date of Performance: 23/02/26

Subject Code: 25CAP-656

1. Aim/Overview of the practical: Convert and Deploy an Employee Management Application using Enterprise Java Components and Apply Validation Techniques

2. Coding:

Index.html file:

```
<!DOCTYPE html>
<html>
<head>
  <title>Authentication</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body class="bg-light">

  <div class="container mt-5">
    <div class="card p-4 shadow">

      <div class="text-center mb-4">
        <button class="btn btn-primary me-2" onclick="showLogin()">Login</button>
        <button class="btn btn-success" onclick="showSignup()">Signup</button>
      </div>

      <form id="loginForm" action="auth" method="post">
        <input type="hidden" name="action" value="login">
        <input class="form-control mb-3" type="text" name="username" placeholder="Username" required>
        <input class="form-control mb-3" type="password" name="password" placeholder="Password" required>
        <button class="btn btn-primary w-100">Login</button>
      </form>

      <form id="signupForm" action="auth" method="post" style="display:none;">
        <input type="hidden" name="action" value="signup">
        <input class="form-control mb-3" type="text" name="username" placeholder="Username" required>
        <input class="form-control mb-3" type="email" name="email" placeholder="Email" required>
```



```
<input class="form-control mb-3" type="password" name="password" placeholder="Password" required>
<button class="btn btn-success w-100">Signup</button>
</form>

</div>
</div>

<script>
function showLogin(){
    document.getElementById("loginForm").style.display = "block";
    document.getElementById("signupForm").style.display = "none";
}
function showSignup(){
    document.getElementById("loginForm").style.display = "none";
    document.getElementById("signupForm").style.display = "block";
}
</script>

</body>
</html>
```

User.java file: to store the user data as json in tomcat directory

```
package model;
public class User {

    private String username;
    private String email;
    private String password;

    public User() {}

    public User(String username, String email, String password) {
        this.username = username;
        this.email = email;
        this.password = password;
    }
    public String getUsername() {
        return username;
    }
    public void setUsername(String username) {
        this.username = username;
    }
    public String getEmail() {
        return email;
    }
```



```
}  
public void setEmail(String email) {  
    this.email = email;  
}  
public String getPassword() {  
    return password;  
}  
public void setPassword(String password) {  
    this.password = password;  
}  
}
```

Auth.java file:

```
package controller;  
import model.User;  
import com.google.gson.Gson;  
import com.google.gson.reflect.TypeToken;  
import java.io.*;  
import java.lang.reflect.Type;  
import java.nio.file.*;  
import java.util.*;  
import jakarta.servlet.annotation.WebServlet;  
import jakarta.servlet.http.*;
```

```
@WebServlet("/auth")
```

```
public class Auth extends HttpServlet {
```

```
    protected void doPost(HttpServletRequest request,  
                           HttpServletResponse response)  
        throws IOException {
```

```
        String action = request.getParameter("action");  
        String username = request.getParameter("username");  
        String password = request.getParameter("password");
```

```
        String filePath = getServletContext().getRealPath("/users.json");
```

```
        Gson gson = new Gson();  
        List<User> userList = new ArrayList<>();  
        File file = new File(filePath);  
        if(file.exists()){  
            String json = new String(Files.readAllBytes(Paths.get(filePath)));  
            Type type = new TypeToken<List<User>>(){}.getType();  
            userList = gson.fromJson(json, type);
```



```
if(userList == null) userList = new ArrayList<>();
}
if("signup".equals(action)){
    String email = request.getParameter("email");
    for(User u : userList){
        if(u.getUsername().equals(username)){
            response.getWriter().println("<h3>User already exists</h3>");
            return;
        }
    }

    User newUser = new User(username, email, password);
    userList.add(newUser);
    Files.write(Paths.get(filePath),
        gson.toJson(userList).getBytes());

    response.getWriter().println("<h3>Signup Successful! Go back and Login.</h3>");

} else if("login".equals(action)){
    for(User u : userList){
        if(u.getUsername().equals(username) &&
            u.getPassword().equals(password)){
            response.sendRedirect("welcome.html");
            return;
        }
    }
    response.getWriter().println("<h3>Invalid Credentials</h3>");
}
}
```

OUTPUT:

Login

Signup

ayush yadav

Login

Invalid Credentials



Signup Successful! Go back and Login.

Welcome! You are logged in!

3. Learning outcomes:

1. Understand client-server interaction in a web application.
2. Implement user authentication using Servlets.
3. Perform server-side form handling using HTTP POST requests.
4. Store and retrieve structured data using JSON files.
5. Apply MVC architecture using model and controller packages.
6. Implement basic input validation and duplicate user checking.